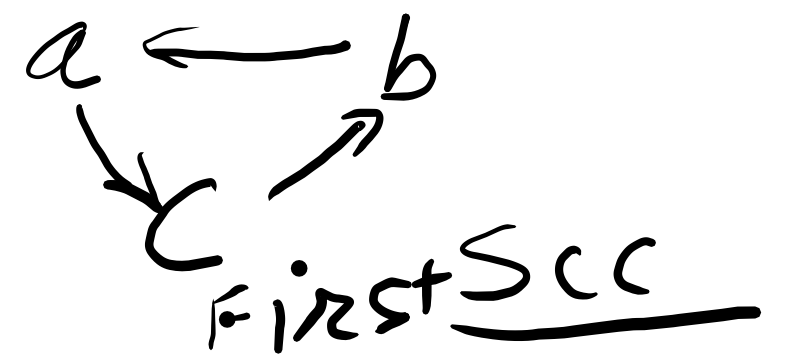
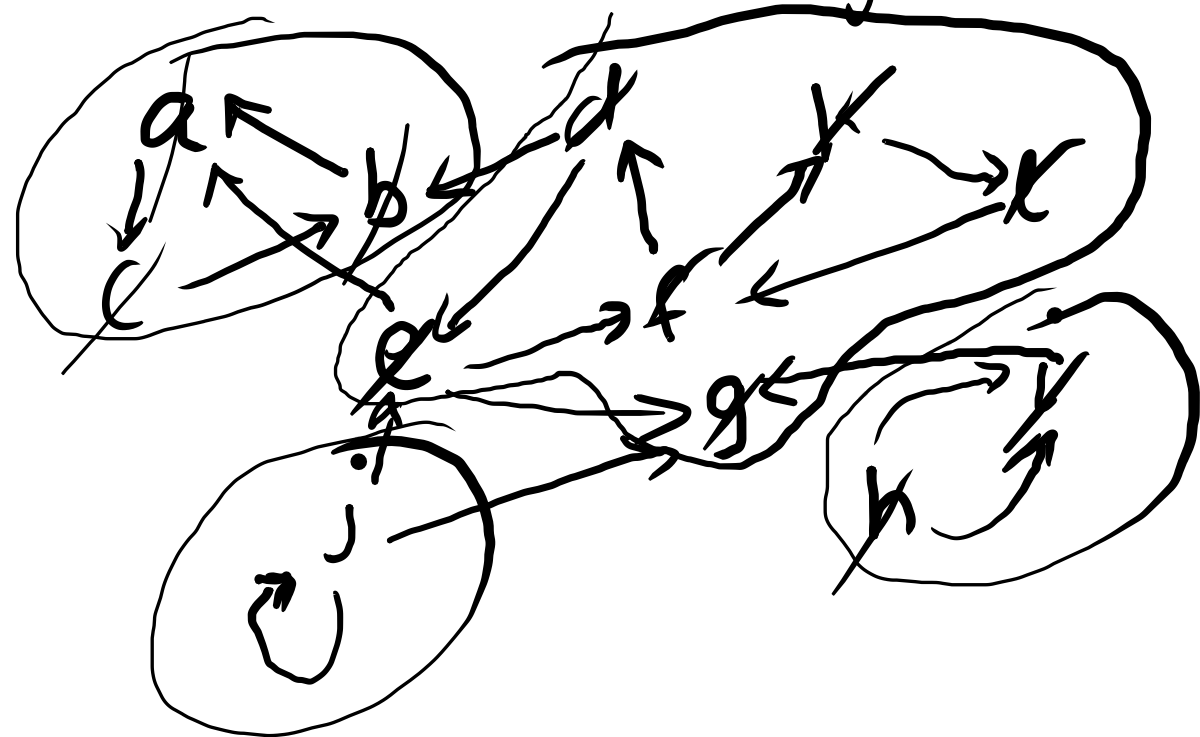


Step 3: Perform DFS on original graph G by obeying the following rules

Rule 1: Start DFS at the first vertex of L

Rule 2: whenever a restart is needed, start from the first vertex of L that is still white.

$L = \{a, b, f, d, e, g, i, k, j, h, c\}$
start DFS from a , which finishes after discovering $\{a, b, c\}$



start DFS from f 2nd SCC
 $\{f, k, e, d, g\}$
start DFS from i 3rd SCC
 $\{i, h\}$
start DFS from j 4th SCC
 $\{j\}$

Time Complexity of Finding SCCs:

Step 1: Reverse the direction of Edges $G \rightarrow G^R$
 $O(|E|)$

Step 2: Perform DFS on G^R to create L^R
 $O(|V| + |E|)$

Obtain L by reversing L^R
 $O(|V|)$

Step 3: Rule 1: Run DFS from 1st vertex in L
 $O(|V| + |E|)$

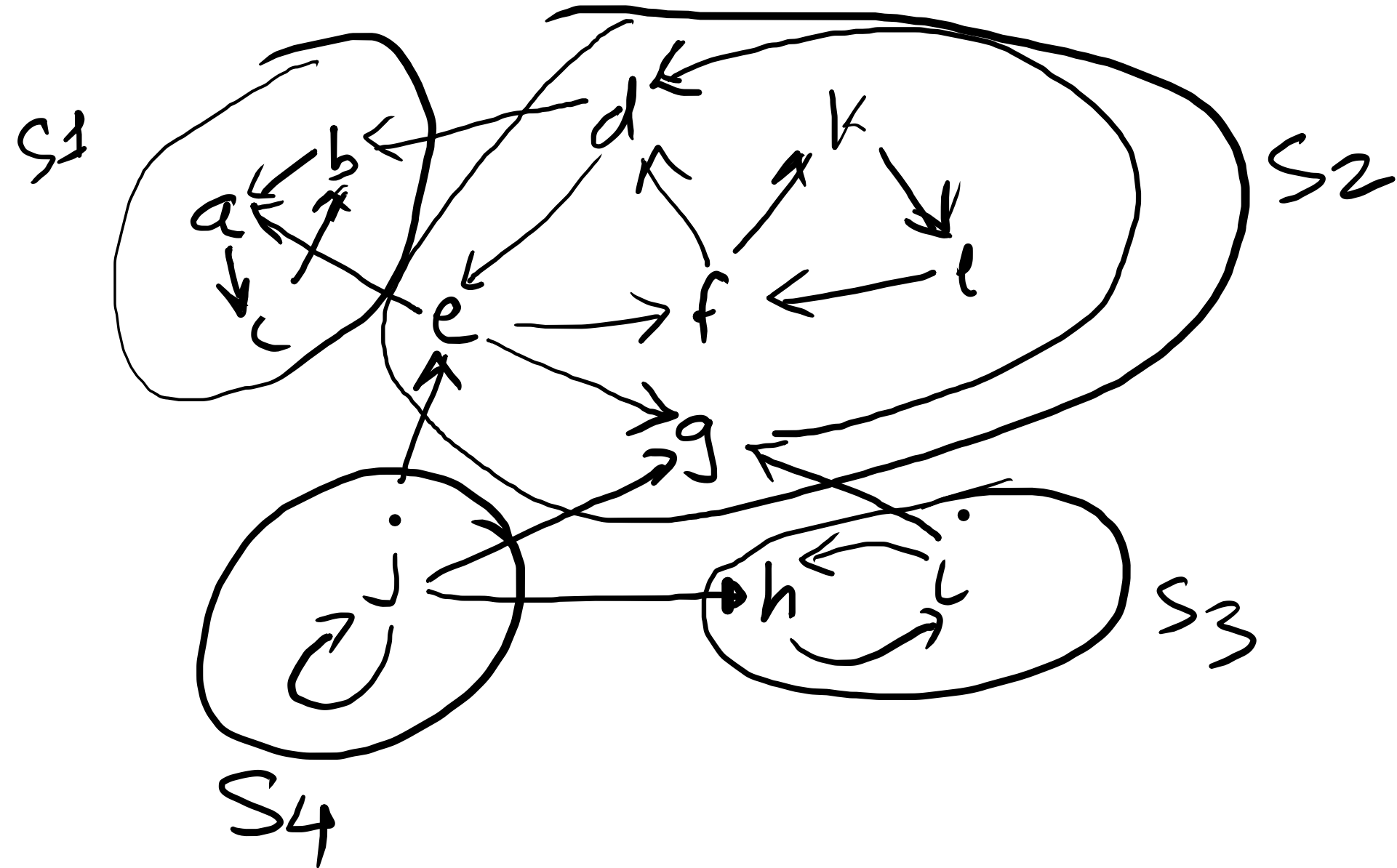
Rule 2: Restart DFS from vertex in L still white
 $N \times (|V| + |E|)$ assuming $N \ll V$
 $O(|V| + |E|)$

$$\begin{aligned}
 & \text{Step 1} + \text{Step 2} && + \text{Step 3} \\
 & O(1E) + O(|V| + |E|) + O(|V|) && + O(|V| + |E|) \\
 \approx & O(|V| + |E|)
 \end{aligned}$$

Proof of correctness:

SCC graph

- Let G be an input directed graph, with SCC $S_1, S_2, \dots, S_t$ for some $t \geq 1$
- Let us define SCC graph G^{SCC} as
 - Each vertex in G^{SCC} is a distinct SCC in G
 - Consider two vertices (i.e. SCCs) S_i and S_j ($1 \leq i, j \leq t$). G^{SCC} has an edge from S_i to S_j if and only if
 - $i \neq j$
 - There is a path in G from a vertex in S_i to vertex in S_j



SCC graph

